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RESEARCH

Superintelligence, instrumental convergence, and the limits of AI apocalypse

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Abstract

This paper examines the metaphysical and technical possibilities of superintelligent artificial agents and their potential to pose existential risks. While Nick Bostrom's Instrumental Convergence Thesis (ICT) suggests that advanced intelligence will systematically adopt power-seeking subgoals, I argue that such risks are less imminent than often portrayed. By distinguishing between metaphysical and technical possibility, I highlight computational constraints such as combinatorial explosion and Moravec's Paradox that limit superintelligence in practice. I further contend that because AI is built "by humans, for humans, about humans", its motivations are likely to remain human-centered in the foreseeable future. Nevertheless, ICT underscores the importance of addressing the conditions under which intelligence and motivation combine to generate risk. I conclude by considering strategies for mitigation, including multi-agent architectures, modular service frameworks, and systems designed with uncertainty about human preferences. Overall, the existential threat of power-seeking superintelligence should be treated as possible but improbable in the short term, warranting caution without succumbing to alarmism.

Keywords Superintelligence · Instrumental convergence · Existential risk · AI alignment · Motivation · Control problem · Bostrom

1 Introduction

The accelerating progress of artificial intelligence (AI) has revived speculation about the emergence of superintelligent agents and the existential risks they may pose. Bostrom [1] famously warned that sufficiently capable systems, regardless of their initial goals, would converge on power-seeking instrumental values such as self-preservation, resource acquisition, and goal-content integrity. This *Instrumental Convergence Thesis* (ICT) underpins the fear that superintelligence could escape human control and threaten civilization. The strength of this concern depends on both metaphysical and physical possibilities [2–4].

Metaphysically, superintelligence may be possible in principle, just as solving combinatorial problems such as the Traveling Salesman Problem is possible "in theory." Technically, however, computational limits and domain-specific constraints make achieving superintelligence intractable, at least in the short term. This paper uses the terms "short term" and "long term" to describe the relative timescales for reaching superintelligence. As Bostrom cautions, assigning specific time frames is difficult because the trajectory toward superintelligence is highly uncertain [4]. Nevertheless, acknowledging that even low-probability, high stakes outcomes warrant nontrivial attention, and thus limited preparation for potential long term risks remains justified. The appropriate stance, however, must be appropriate—allocating effort relative to the expected likelihood of emergence rather than assuming imminence [4, 5]. Similarly, while ICT may hold at a conceptual level, the practical development of AI reflects human-centered design: systems are built by humans, for humans, and about humans. From training data to reinforcement learning from human feedback, contemporary AI embeds human motivations and values, suggesting that orthogonal, alien goal systems are less likely to emerge in practice. Moreover, accounts from physics suggest that intelligence may emerge by leveraging environmental structure and entropy gradients rather than by brute-forcing optimal computation [6, 7]. Greater

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capability does not eliminate complexity limits but instead uses patterns to operate more efficiently within them. Thus, superintelligent agents may achieve higher performance through heuristics and structure-exploiting strategies without violating fundamental computational constraints [8].

This paper evaluates ICT in light of both its philosophical appeal and its technical challenges. I argue that existential risk from superintelligence remains low-probability in the short term but merits philosophical and technical analysis given the potential stakes. I further consider mitigation strategies, including distributed architectures, uncertainty-based alignment methods, and frameworks such as Drexler's Comprehensive AI Services (CAIS). My aim is not to dismiss superintelligence risk, but to reframe it as one possibility among many, requiring careful probabilistic reasoning and practical safeguards.

2 Technical and metaphysical possibility

Karl Menger's Traveling Salesman Problem involves finding the shortest possible route that a salesperson can take to visit a given number of cities exactly once and return to the starting city [9]. Metaphysically, the problem is possible and straightforward. An algorithm that checks all tours will eventually find the solution, which can be checked quickly. However, technically finding the solution is *intractable*—that is, it belongs to a class of problems whose running time grows super-polynomially with input size, as formalized in computational complexity theory [10, 11]. The Traveling Salesman Problem is a P vs. NP problem, meaning that while a solution can be verified quickly, no known algorithm can find it efficiently. Even a hypothetical superintelligence constrained by the same physical laws would face such exponential scaling limits in search and planning. For P vs. NP problems, there is no known way to find an answer quickly, including for quantum computers or superintelligence; but this does not imply that useful performance requires overcoming such limits. In practice, intelligent systems, including human cognition, rely on heuristics and computational rationality to trade accuracy for time and memory, achieving good enough solutions without exhaustive search [5, 10, 12]. The short term constraint is not logical impossibility but rate and scale under resource bounds, with heuristics enabling competence short of combinatorial optimality. Understanding metaphysical possibilities, like superintelligence, may be easy. In principle, it is easy to solve the Traveling Salesman Problem and understand superintelligence's capabilities. However, understanding *technical* possibilities—what can be achieved within physical and computational limits—is also important in understanding short term and some long term risks of superintelligence.

Michael Wooldridge, for example, describes that rapid cognitive enhancement in superintelligence is "deeply implausible" because of computational limits, such as combinatorial explosion 13. However, Bostrom uses Pascal's Mugging to suggest that giving superintelligence a flawed decision theory could be disastrous. Bostrom uses Bayesian reasoning to estimate the probability of these risks. Bayesian reasoning suggests spending some nontrivial amount of effort considering existential risk because it is low-probability but high stakes. Yet Bostrom ultimately claims he is optimistic about the outcome of superintelligence, despite high stakes [14]. Spending effort on too unlikely scenarios may be irrational. I will argue that the existential risk of power-seeking behavior in AI is low, at least in the short term, and therefore does not warrant significant concerns.

3 The orthogonality thesis reconsidered

Artificial agents may have an enormous range of possible final goals. Bostrom claims that AI can be far less human-like in its motivations than a space alien. One can easily conceive of an AI whose sole goal is to calculate decimal places of pi indefinitely. Bostrom claims that it would be easier to create an AI with such a simple goal like calculating decimal places of pi, than to build one that has a human-like set of values and dispositions. I agree that this claim may be true in principle, but I will argue that in practice, AI will likely adopt human-like motivations and values.

4 Human influence on AI motivation

Current AI models are made by humans, for humans, about humans. For example, ChatGPT is trained by humans and reinforced with human feedback (RLHF) to fit human goals (e.g., helpful, harmless, and honest), for human users, and its training corpus is language (a human artifact) [15]. However, this dependence on human data and design does not guarantee moral alignment. Human-made systems, such as companies and governments, can also produce harmful or unintended outcomes despite their human origin. Likewise, future artificial superintelligence may not necessarily rely on present-day methods such as reinforcement learning from human feedback or supervised training on human data, limiting the generalizability of this claim. Therefore, while current AI systems are likely to remain human-centered in design and motivation, this expectation is probabilistic rather than absolute, and future architectures could diverge from human values.

I argue that AI must have a first cause, an action, which is human-motivated. AI is human-made intelligence. In *A*

Treatise of Human Nature, David Hume claimed that "reason is, and ought only to be the slave of the passions" [9]. Reason alone does not motivate; passions or desires provide motivation. In that sense, motivation causes action. Because the first cause of AI is human, AI is influenced by human motivation through design choices. The data, selection of data, and objective functions AI is trained on reflect human beliefs, values, and motivations. For example, take ChatGPT. Its main data source is human-generated content on the internet. ChatGPT will also often incorporate common falsehoods in the training data, which is evidence that beliefs reflect training data [16]. Choices about what training data to include or exclude, how to weight or filter it, and how to label preference data for fine-tuning express values (i.e., what we think matters) and morals (i.e., what should or should not be done). The extent to which ChatGPT refuses to assist with harmful instructions is a human-made moral decision. Google's Gemini image generation was once trained to show diverse people, which produced some inaccurate images [17]. Optimizing for diversity can compromise truth, a clear value trade-off [18]. Current training methods such as reinforcement learning from human feedback, data-protection oversight, and constitutional AI all reflect human beliefs, values, and motivations.

However, as Dennett [19] warns against the "Philosopher's Disease"—mistaking a failure of imagination for an insight into necessity—it would be an error to assume that human beings will always remain the sole constitutive agents in AI systems. Future artificial agents could acquire partial autonomy in goal-seeking and develop something like a will of their own as their architectures become more self-modifying [20]. Moreover, Elena Popa [21] argues that while current AI goals are tied to human purposes, this connection may weaken as AI gains the capacity to redefine or generate objectives independent of human oversight. Thus, my claim that AI is "by humans, for humans, about humans" should be understood as an empirical observation about present-day design, not as a metaphysical necessity. Recognizing this contingency strengthens, rather than weakens, the argument by clarifying that human motivation currently shapes AI development while leaving open the analytic possibility of future autonomous agency.

5 Instrumental convergence thesis and risk conditions

In *The Superintelligent Will*, Bostrom describes ICT: "several instrumental values can be identified which are convergent in the sense that their attainment would increase the chances of the agent's goal being realized for a wide range of final goals and a wide range of situations, implying that these instrumental values are likely to be pursued by many intelligent agents" [22]. The thesis contains three broad qualifiers: "wide range of final goals," "wide range of situations," and "likely to be pursued." If AI may pose an existential risk, then it is important to thoroughly understand the solution space defined by these qualifiers. Which set of final goals would or would not cause an agent to attain instrumental values? Which set of situations would or would not cause an agent to attain instrumental values? Under what circumstances would or would not an intelligent agent pursue instrumental values?

Is it technically easier to build a superintelligence that optimizes counting digits of pi rather than human welfare or moral goodness? Counting pi to arbitrary digits needs no superintelligence." The most advanced AI today—ChatGPT, AlphaFold—was funded because it solved economically valuable or scientifically interesting *human* problems. Innovation follows problem pressure and economic returns (usually human-centered tasks). Therefore, investors, governments, and researchers often chase problems in health, finance, and defense. Training data are mostly human language, images, code. Community prestige attaches to beating ImageNet, solving protein folding, winning at Go, all of which are human-defined challenges.

Bostrom points out that simple goals like counting digits of pi are easier to specify than human goals like "happiness." However, working under Bostrom's assumption of functionalism, it is metaphysically possible to quantify happiness or other complex human goals [3]. The question then becomes: What will we technically achieve first: specifying happiness or building superintelligence that understands mathematics, literature, politics, and physics well enough to be *dangerous*?

I argue that for a superintelligent agent to pose existential risk by the ICT, the agent must satisfy *both* the intelligence and motivation conditions. Importantly, bounded-rational heuristics may be sufficient for convergent instrumental behaviors to emerge in practice, even absent exact optimization [5, 10]. Absent just one of these conditions, the agent may pose substantially less of an existential risk. In *Superintelligence: Paths*, *Dangers*, *Strategies*, Bostrom presents the control problem: How do humans know the superintelligent agent is acting in humanity's interest? [4]. To control an agent's capability, Bostrom outlines boxing, incentives, stunting, and tripwires. To select an agent's motivation, Bostrom outlines direct specification, domesticity, indirect normativity, and augmentation. Ideally, humans solve the control problem both for capability and motivation. However, it may be the case that solving the control problem for either capability or motivation may be sufficient. An agent with the motivation to destroy the world but no capability is not capable of destroying the world. An agent with no

motivation to destroy the world but has the capability will not destroy the world. Therefore, if we can solve the control problem for either the intelligence or motivation condition before creating superintelligent agents, then we can limit the risk of these agents.

6 Limits of superintelligence in the natural world

Another metaphysical limitation is time. Imagine a superintelligent agent that has both the capability and motivation to destroy the world but lacks sufficient time. Time may limit the effects of superintelligent agents. I do concede to Bostrom's point that speed superintelligence might run on a radically faster "clock" than ours [4]. However, there is a distinction between faster intelligence and faster actions. Current AI has fast intelligence in some domains. For example, ChatGPT can read a book in a few seconds and write a PhD thesis in an afternoon. Yet AI seems to lag behind in perception and in motor skills. Moravec's Paradox, for instance, describes the observation that tasks easily performed by humans, like perception and motor skills, are difficult for AI, while tasks that are difficult for humans, such as complex calculations, are relatively easy for machines [23]. Time can limit an agent's intelligence and actions independently. That is, an agent can (and to an extent already does) have fast intelligence but lacks fast actions in the natural world.

However, as contemporary AI infrastructure demonstrates, time constraints on single agents can be mitigated by distributed computation—where updates and knowledge propagate across multiple systems almost instantaneously. This "pooled timeline" of collective processing partially offsets temporal limitations at the individual level [24]. Nonetheless, such scaling does not erase physical and causal constraints: even massively parallel systems remain bound by bandwidth, coordination overhead, and environmental interaction speed. Therefore, while distributed architectures accelerate reasoning, physical time still bounds the realization of instrumental goals in the natural world.

In *The Superintelligent Will*, defines intelligence as "the capacity for instrumental reasoning" [3]. ICT assumes that once an agent is sufficiently capable, it will tend to adopt subgoals. ICT treats "capability" as if reasoning is enough to achieve subgoals. However, take the subgoals of self-preservation and resource acquisition for example. Self-preservation only has value when the agent has knowledge of itself (hardware and software), and resource acquisition only has value when the agent has knowledge of the world. For example, ChatGPT may be intelligent, but it does not have illegal or confidential knowledge. Therefore, intelligence and knowledge are (to an extent) orthogonal, and ICT seems to only hold for agents that are *both* intelligent and sufficiently knowledgeable. I would agree with Bostrom's likely reply that an intelligent agent would seek knowledge. However, seeking knowledge itself has a cost (e.g. time and compute). This idea seems also to inspire Bostrom's capability control methods in the control problem (e.g. limiting internet access).

7 Mitigating power-seeking risks

The ICT assumes one superintelligent agent with sufficient capability will pursue instrumental goals. However, multiple agents, each with insufficient capability to pursue instrumental goals, could still collectively possess high capability. Multiple agents will likely be collectively even more capable than a single agent because of the potential advantages of cooperative distributed problem solving [25]. Many agents can work together to solve problems that are beyond their individual capabilities.

Multi-agent systems also align with Eric Drexler's Comprehensive AI Services (CAIS), a modular ecosystem of *task-oriented* services that users compose on demand [26]. For example, ChatGPT uses this framework by calling distinct tools (e.g. browser or code interpreter), and each tool runs in its own environment and is invoked only when needed.

Lastly, Stuart Russell suggests making AI "provably beneficial" through setting uncertain objectives [27]. If an AI's only objective is to realize human preferences and is initially uncertain about what those preferences are, then it must learn about them from human behavior. This AI may reduce the effect of Bostrom's idea of perverse instantiation: superintelligence satisfying its final goal but violating a programmer's intention. AI with uncertain objectives may better understand a programmer's intention. For example, if an AI's final goal were to "make us smile," it might ask follow-up questions, ultimately understanding that the programmer's intention was something like having the AI tell a joke.

8 Conclusion

Superintelligence and its associated risks occupy a paradoxical space between metaphysical possibility and technical possibility. Bostrom's ICT offers a compelling philosophical framework: if intelligence entails convergent instrumental goals, then sufficiently capable agents may pursue power in ways dangerous to humanity. Yet, as I have argued, technical constraints, human-centered design practices, and

the distinction between intelligence and knowledge substantially diminish the likelihood of catastrophic outcomes in the short term.

The existential threat of power-seeking AI is real enough to demand attention, but not so pressing as to overshadow more immediate challenges in AI ethics, governance, and design. Risk mitigation strategies such as multi-agent architectures, modular service ecosystems, and alignment via uncertain objectives offer promising avenues for reducing long term dangers without succumbing to speculative alarmism. In this way, we can take Bostrom's warnings seriously while grounding them in a balanced assessment of both technical possibility and metaphysical possibility.

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Declarations

Competing interests The authors declare no competing interests.

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